

OpenCPU SDK

GNSS API Notes

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Notice

This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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Sales@neoway.com

Support@neoway.com

Website: <http://www.neoway.com>

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About This Document

Scope

This document is applicable to N706B and N520 modules.

Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Dong Nengxiang

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

This document describes the API related to the Location operation in the OpenCPU SDK.

2 Data Structure

2.1 nwy_loc_startup_mode_e

```
typedef enum
{
    NWY_LOC_HOT_START = 0,
    NWY_LOC_WARM_START = 1,
    NWY_LOC_COLD_START = 2,
    NWY_LOC_FACTORY_START = 3,
} nwy_loc_startup_mode_e;
```

2.2 nwy_loc_output_type_e

```
typedef enum
{
    GNSS_OUTPUT_GGA = 0,
    GNSS_OUTPUT_GSA = 1,
    GNSS_OUTPUT_GSV = 2,
    GNSS_OUTPUT_RMC = 3,
    GNSS_OUTPUT_VTG = 4,
    GNSS_OUTPUT_GLL = 5,
    GNSS_OUTPUT_ZDA = 6,
    GNSS_OUTPUT_ALL = 99,
} nwy_loc_output_type_e;
```

2.3 nwy_loc_rate_e

```
typedef enum
{
    NWY_LOC_1HZ = 0,
    NWY_LOC_2HZ = 1,
} nwy_loc_rate_e;
```

2.4 nwy_gnss_output_info_ind_t

```
typedef struct
{
    uint16_t length;
    uint16_t type;
    uint8_t nmea_data[4096];
}
```

```
}nwy_gnss_output_info_ind_t;
```

2.5 nwy_setserver_para_t

```
typedef struct  
{  
    int set_flag;  
    int data_type;  
    char url[521+1];  
    char user[128];  
    char pass[128];  
    char mid[64];  
    int port;  
  
    double lat;  
    double lng;  
    double alt;  
}nwy_setserver_para_t;
```

3 APIs

The API definitions can be found in `nwy_loc_api.h`, and is used to implement positioning-related functions.

3.1 `nwy_loc_navigation_start`

Function	<code>nwy_error_e nwy_loc_navigation_start(void)</code>
Description	Enables the GNSS function.
Parameter	void
Return Value	Successful: <code>NWY_SUCCESS</code> Failure: Other enumerated values
Notes	NA

3.2 `nwy_loc_navigation_stop`

Function	<code>nwy_error_e nwy_loc_navigation_stop(void);</code>
Description	Disables the GNSS function.
Parameter	void
Return Value	Successful: <code>NWY_SUCCESS</code> Failed: another enumerated value

3.3 `nwy_loc_mode_set`

Function	<code>nwy_error_e nwy_loc_mode_set (int mode)</code>
Description	Sets the positioning mode.
Parameter	mode: Positioning mode. See <code>nwy_loc.h</code> for definitions: 0:GPS 1:BD 2:GPS+BD 3:GPS+GLONASS

	4:GPS+GALILEO 5:BD+GLONASS 6:BD+GALILEO 7:GLONASS+GALILEO 8:GLONASS 9:GALILEO 10:GPS+BD+GLONASS 11:GPS+BD+GALILEO 12:GPS+GLONASS+GALILEO 13:BD+GLONASS+GALILEO 14:GPS+BD+GLONASS+GALILEO
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value
Notes	Supported modes depend on the chip specifications. Default supported modes for this chip are (0, 1, 2, 3, 4). Default positioning mode: GPS + BD + QZSS (QZSS is included in GPS by default).

3.1 nwy_loc_startup_set

Function	nwy_error_e nwy_loc_startup_set(nwy_loc_startup_mode_e startup)
Description	Sets the positioning startup mode.
Parameter	Startup: Startup mode 0: hot start 1: Warm start 2: Cold start
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value
Notes	NA

3.2 nwy_loc_rate_set

Function	nwy_error_e nwy_loc_rate_set(nwy_loc_rate_e rate);
Description	Sets the positioning rate.
Parameter	rate: location rate 0: 1HZ 1: 2HZ
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value

Notes	The GNSS chip defaults to 1Hz and can be configured to 2Hz.
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3.3 nwy_loc_nmea_get

Function	nwy_error_e nwy_loc_nmea_get(int type,char *data);
Description	Sets the positioning startup mode.
Parameter	type: Type of NMEA data to retrieve: GGA sentence GSA sentence GSV sentence RMC sentence VTG sentence GLL sentence All NMEA sentences data: Buffer to store the retrieved NMEA sentences.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value
Notes	NA

3.4 nwy_loc_recv_cb_reg

Function	nwy_error_e nwy_loc_recv_cb_reg(nwy_loc_ind_event_func ind_handler);
Description	Registers a callback function for NMEA sentence reception.
Parameter	typedef void (*nwy_loc_ind_event_func)(nwy_gnss_output_info_ind_t *ind_msg); // callback function typedef struct { uint16_t length; uint16_t type; uint8_t nmea_data[4096]; }nwy_gnss_output_info_ind_t;
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value
Notes	NA

3.5 nwy_loc_rcv_cb_del

Function	nwy_error_e nwy_loc_rcv_cb_del(void);
Description	Unregisters the NMEA callback function.
Parameter	void
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value
Notes	NA

3.6 nwy_loc_agNSS_start

Function	nwy_error_e nwy_loc_agNSS_start(nwy_setserver_para_t *server_info);
Description	Configures the AGPS server and retrieves assistance data for injection into the GNSS chip.
Parameter	<pre>t typedef struct { int set_flag; int data_type; char url[521+1]; char user[128]; char pass[128]; char mid[64]; int port; double lat; double lng; double alt; }nwy_setserver_para_t;</pre>
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value
Notes	This function must be called after enabling the GNSS interface. Set correct AGNSS server credentials. Test accounts are available for demo purposes; official accounts must be requested from Neoway.